

Sample Passport

Sentinel Vancomycin Precision-Dosing Cohort True-Subset Sample — 400 Patients

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Version 1.0 | Stage 5 Delivery

1. Sample Identification

This passport documents the true-subset sample of the Sentinel Vancomycin Precision-Dosing Cohort. The sample contains 400 patients drawn from the full 10,000-patient dataset via stratified sampling. All rows in the sample are byte-identical to the corresponding rows in the full dataset — this is a verified true subset, not a re-simulation.

2. Sample Statistics

The sample is stratified across the four subpopulations to ensure representation of all phenotypes present in the full cohort.

Property	Value	Description
Patients	400	Stratified from 10,000
Dosing regimens	1,200	3 per patient
PK profile rows	1,038,000	865 time points per regimen
Sample size vs full	4.0%	400 / 10,000
File format	Parquet (Snappy)	Same as full dataset

3. Phenotype Coverage

All phenotypes from the full dataset are represented in the sample, including special populations (RRT, ARC, MRSA) and the full MIC distribution.

Phenotype	Count	Proportion	Covered
Normal	142	35.5%	Yes
Obesity	104	26.0%	Yes
Elderly (>=65)	95	23.8%	Yes
ICU	59	14.8%	Yes
RRT (within ICU)	9	2.3%	Yes
ARC (within ICU)	35	8.8%	Yes
MRSA	143	35.8%	Yes
MIC 0.5 mg/L	40	10.0%	Yes
MIC 1.0 mg/L	265	66.3%	Yes

Phenotype	Count	Proportion	Covered
MIC 2.0 mg/L	79	19.8%	Yes
MIC 4.0 mg/L	16	4.0%	Yes
AUC/MIC < 400	104	26.0%	Yes
AUC/MIC 400-600	68	17.0%	Yes
AUC/MIC > 600	228	57.0%	Yes
Trough < 10 mg/L	91	22.8%	Yes
Trough > 20 mg/L	225	56.3%	Yes

4. Verification Results

Seven independent checks were performed to verify that the sample is a true subset of the full dataset. All checks passed.

Check	Result
1. Patient IDs exist in full dataset	PASS
2. Regimen IDs exist in full dataset	PASS
3. Patient rows byte-identical to full	PASS
4. Regimen rows byte-identical to full	PASS
5. Profile rows byte-identical to full	PASS
6. Proper subset (no extra rows)	PASS
7. No null values in any column	PASS

Verification: All 7 checks passed. Patient rows, regimen rows, and profile rows are byte-identical to the full dataset. The sample is a verified true subset.

5. Sample Files

The sample is delivered as three Parquet files, mirroring the structure of the full dataset.

File	Rows	Columns	Size (MB)
vanco_sample_patients.parquet	400	29	0.08
vanco_sample_dosing_regimens.parquet	1,200	17	0.13
vanco_sample_pk_profiles.parquet	1,038,000	6	17.50

6. Key Pharmacokinetic Formulas

The sample is generated from the same two-compartment model as the full dataset. Key equations are shown below.

Eq. 1 Two-compartment mammillary ODE system

$$\frac{dA_1}{dt} = -\frac{CL}{V_1} A_1 - \frac{Q}{V_1} A_1 + \frac{Q}{V_2} A_2 + R(t)$$

$$\frac{dA_2}{dt} = \frac{Q}{V_1} A_1 - \frac{Q}{V_2} A_2$$

$$\frac{dA_{elim}}{dt} = \frac{CL}{V_1} A_1$$

Two-compartment ODE system (central, peripheral, elimination).

Eq. 4 Steady-state 24-hour AUC

$$AUC_{24,ss} = \frac{\text{Daily Dose}}{CL_{total}}$$

Steady-state 24-hour AUC = Daily Dose / CL.

Eq. 5 Clearance covariate model (Thomson 2009 + Colin 2019)

$$CL = \theta_{CL} \left(\frac{WT}{70} \right)^{0.75} (1 + \theta_{CrCL} (CrCL - 66)) (1 + \theta_{ARC} I_{ARC}) \exp(\eta_{CL})$$

Individual clearance with covariates and IIV.

Eq. 8 Combined proportional + additive residual error

$$C_{obs} = C_{pred} (1 + \sigma_{prop} \varepsilon_1) + \sigma_{add} \varepsilon_2$$

$$\varepsilon_1, \varepsilon_2 \sim \mathcal{N}(0, 1), \quad \sigma_{prop} = 0.15, \quad \sigma_{add} = 1.6 \text{ mg/L}$$

Combined proportional + additive residual error.

7. Relationship to Full Dataset

The sample is a 4% stratified subset of the full 10,000-patient cohort. It is intended for preliminary analysis, method development, and evaluation before committing to the full 450 MB dataset. All results derived from the sample are directly comparable to the full dataset because the data is byte-identical.

8. Limitations

The sample is small (n=400) and may not capture rare parameter combinations. For definitive analysis, the full dataset should be used. The sample is provided as a convenience for initial exploration and benchmarking.

9. Bibliography

All 26 references used in the full dataset apply equally to this sample. See the full data passport for the complete bibliography.

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